Assignment 2: Business Process Analysis and Redesign

IT Helpdesk: As-Is Analysis, Staffing and To-Be Design

INFO-H420 — Process Modeling, Data Pipelines, and Responsible AI

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Abstract

This report analyzes the supplied IT-helpdesk process and presents a possible redesign. The analytical as-is cycle-time efficiency is 4.742%, with a processing labor cost of €176.67 per execution under Exercise 1's handoff exclusion. An Erlang-C calculation requires three Level-2 employees for a two-hour resolution service at one request per hour; including evaluation and writing in a single service visit requires four. At the four-request-per-hour workload of Exercise 3, three Level-2 employees are insufficient. A 12-configuration simulation identifies the capacity bottleneck and diminishing returns beyond approximately 12 Level-2 employees. Six issues are linked to redesign proposals. The selected to-be model automates routine delivery and feedback routing while retaining human assistance. At matched staffing, the main redesign run reduces modeled labor cost per case by approximately 7%; its smaller cycle-time change requires more cautious interpretation.

Scope and modeling conventions

The assignment describes five Level-1 employees at €40 per person-hour and three Level-2 employees at €60 per person-hour, serving approximately 500 company employees [1]. Each staffed activity requires *one* employee. The capacity of three is a shared pool, not a three-person team per request.

The supplied BPMN model is used for Exercises 1 and 3. Known requests occur with probability 0.2; otherwise the request is escalated. Each delivered Level-2 resolution succeeds with probability 0.8. Unsuccessful cases return to resolution work, not to initial evaluation. Repeated outcomes are modeled as independent with unchanged probabilities. No business-hour calendar is imposed because the assignment provides no shift schedule. All reported hours are continuous modeled hours unless explicitly identified as working hours in the queueing calculation.

1 Exercise 1: cycle-time efficiency and cost per execution

1.1 Expected activity visits

Let $p = 0.8$ denote the probability of escalation and $r = 0.2$ the probability that a resolution fails. Conditional on escalation, the number of resolution attempts N is geometrically distributed:

$$\mathbb{E}[N \mid \text{escalated}] = \frac{1}{1-r} = 1.25.$$

Consequently, each activity in the resolution–delivery–feedback loop is visited $p/(1-r) = 1$ time per incoming request on average. Initial evaluation is visited 0.8 times; forwarding unsuccessful feedback is visited $pr/(1-r) = 0.2$ times. The known-resolution branch is visited 0.2 times.

Exercise 1 explicitly excludes waiting and handoffs from processing time and cost. Accordingly, the two-minute forwarding step is excluded from labor but retained as elapsed handoff time. In BpmnGo this is represented by zero processing time and a two-minute unresourced delay. This is an accounting convention for Exercise 1, not a claim that forwarding physically requires no work.

Table 1: As-is inputs and expected visits. Durations are minutes per visit.

Activity	Resource	Work	Delay	Visits/case
Check request	Level 1	10	60	1.0
Send known resolution	Level 1	5	0	0.2
Evaluate request	Level 2	20	120	0.8
Solve request	Level 2	120	1200	1.0
Write resolution	Level 2	20	0	1.0
Send written resolution	Level 1	20	1200	1.0
Receive client response	None	0	1200	1.0
Forward unsuccessful feedback	Handoff	0	2	0.2

1.2 Processing time and cycle time

Expected actual-work time per request is

$$\begin{aligned} T_{L1} &= 10 + 0.2(5) + 1(20) = 31 \text{ min}, \\ T_{L2} &= 0.8(20) + 1(120) + 1(20) = 156 \text{ min}, \\ T_{\text{work}} &= 31 + 156 = 187 \text{ min}. \end{aligned}$$

Expected waiting and elapsed handoff time is

$$T_{\text{delay}} = 60 + 0.8(120) + 1200 + 1200 + 1200 + 0.2(2) = 3756.4 \text{ min}.$$

The configured 20-hour waits before resolution work and delivery apply on each visit through the corresponding loop activities. Thus,

$$T_{\text{cycle}} = 187 + 3756.4 = 3943.4 \text{ min} = 65.723 \text{ h}, \qquad \text{CTE} = \frac{187}{3943.4} \times 100 = \boxed{4.742\%}.$$

The efficiency numerator follows the assignment's definition of actual work. BpmnGo's value-added flags are configured to reproduce that definition; no separate customer-value classification is inferred.

1.3 Cost per execution

Only actual processing labor is charged:

$$C = \frac{31}{60}(40) + \frac{156}{60}(60) = 20.6667 + 156 = \boxed{\text{€176.67 per request}}.$$

This is not the cost of employing the whole team for the request's elapsed cycle time. The BpmnGo analytical calculation reproduced the values above. Given means are sufficient for this calculation, so mean values were entered as fixed durations.

2 Exercise 2: Level-2 staffing for a waiting-time target

2.1 Main queueing model

The completed capacity calculation treats research and resolution as a single Level-2 service station with mean service time two working hours. Requests reaching this station arrive at rate $\lambda = 1$ per working hour. An $M/M/c$ (Erlang-C) model assumes Poisson arrivals, independent exponential service durations, identical employees and a shared FIFO queue, without abandonment. These distributional assumptions are additional to the assignment's stated means.

The service rate per employee is $\mu = 1/2 = 0.5$ requests per hour. For c employees, let $a = \lambda/\mu = 2$ and $\rho = a/c$. A stationary queue requires $\rho < 1$. The Erlang-C probability of waiting and mean queueing wait are

$$P_W = \frac{a^c/[c!(1-\rho)]}{\sum_{n=0}^{c-1} a^n/n! + a^c/[c!(1-\rho)]}, \qquad W_q = \frac{P_W}{c\mu - \lambda}.$$

With $c = 2$, utilization is 100% and no finite steady-state mean wait exists. With $c = 3$, $\rho = 2/3$, $P_W = 4/9$, and

$$W_q = \frac{4/9}{1.5 - 1} = \frac{8}{9}\ \text{h} = 53.33\ \text{min} < 2\ \text{h}.$$

Therefore, **three Level-2 employees suffice under the two-hour resolution-station interpretation**. The target concerns waiting before service, not service plus waiting; the latter is 2.889 hours at $c = 3$.

Table 2: Erlang-C results for one arrival per working hour.

Service boundary	Service time (h)	Minimum staff	Mean wait (h)
Research/resolution only	2.000	3	0.889
Evaluation + resolution + writing, one visit	2.667	4	0.757

2.2 Sensitivity to the service boundary

If "Level-2 service" includes the initial 20-minute evaluation and 20-minute writing task, one visit lasts $(20 + 120 + 20)/60 = 2.667$ hours. With the same $M/M/c$ approximation, three staff then yield $W_q = 6.380$ hours; four yield 0.757 hours. This alternative excludes repeat visits. If one request per hour means *new* requests only, with rework generating additional visits, the arrival/workload model must also account for that feedback. The two staffing answers therefore correspond to different explicit service boundaries; the assignment's mean times alone do not uniquely specify a queueing model.

3 Exercise 3: simulation and resource experiments

3.1 Experiment design

BpmnGo's discrete-event simulator was configured with constant Poisson demand at four requests per hour, a 720-hour admission horizon, 20 replications, and initial seed 42 (replication seeds 42–61). The system starts empty; no warm-up observations are discarded. Resources are continuously available, use a common queue per pool, and are released between activities. Individual staff identities are not retained across a case's successive tasks.

All fixed internal waiting times in Table 1 are set to zero. Only the client-response delay remains, as an unresourced 1200-minute wait. Waiting for an occupied staff pool is generated by the simulation, not inserted as another fixed delay. Forwarding unsuccessful feedback is modeled here as two minutes of Level-1 work. Exercise 1's special handoff exclusion is not applied to this operational simulation.

The known-resolution communication time follows a normal distribution with mean 5 and standard deviation 3 minutes; post-writing communication uses mean 20 and standard deviation 5. Other processing durations are fixed at the supplied means. BpmnGo resamples negative normal draws, so the realized mean of the short activity is slightly higher than the nominal five minutes. This limitation also explains small differences between simulated and analytical labor costs.

The baseline uses 5 Level-1 and 3 Level-2 staff. The factorial campaign compares Level-1 capacities $\{3, 5\}$ and Level-2 capacities $\{3, 6, 9, 11, 12, 15\}$: 12 configurations, each with 20 replications. Two separate detailed runs retain the baseline and the 5/12 configuration for resource-pressure comparison.

3.2 Measured results

The baseline mean cycle is 1063.69 hours, with an approximate 95% confidence interval of 1047.12–1080.26 hours across replication means. Level-2 utilization is 97.37%, its mean queue is 1166.92 requests, and the mean of per-replication maximum queues is 1935. At 5/12, mean cycle falls to 23.81 hours (95% interval 23.68–23.93 hours), Level-2 utilization is 78.32%, and its mean queue falls to 2.25 requests. These metrics include the period after arrivals stop while the remaining workload is drained.

Table 3: Staffing experiment. Cycle times are hours; budget assumes the whole team is paid continuously.

L1	L2	Mean cycle	p95 cycle	Cases/h	Labor €/case	Budget €/h
3	3	1,056.90	2,002.65	1.13	176.91	300
3	6	316.63	715.45	2.20	176.41	480
3	9	82.12	201.04	3.13	176.63	660
3	11	25.69	54.71	3.61	177.22	780
3	12	23.89	48.43	3.58	177.09	840
3	15	23.39	46.74	3.59	177.16	1,020
5	3	1,063.69	2,009.24	1.12	177.26	380
5	6	321.12	720.36	2.18	177.55	560
5	9	85.07	207.99	3.17	177.54	740
5	11	25.42	54.14	3.63	177.06	860
5	12	23.81	48.23	3.61	177.36	920
5	15	23.15	46.19	3.61	176.69	1,100

3.3 Interpretation and trade-offs

The bottleneck can also be established analytically. Expected Level-2 work per incoming request is 156 minutes, so at four arrivals per hour the offered workload is

$$a_{L2} = 4 \times \frac{156}{60} = 10.4 \text{ staff-hours per hour.}$$

Three, six and nine Level-2 employees cannot sustain this workload. Eleven is the smallest integer capacity strictly above it, but has little spare capacity. The results show diminishing returns after approximately 12 Level-2 staff: with five Level-1 staff, increasing Level-2 capacity from 12 to 15 saves about 39 minutes of mean cycle time while increasing the hourly staffing budget from €920 to €1100.

Client feedback becomes dominant once staff queues are small. The expected number of feedback waits per incoming request is $0.8/(1 - 0.2) = 1$, contributing 20 hours. With approximately 3.1 hours of work, the no-resource-queue analytical reference is near 23.1 hours. Merely adding more employees cannot remove this external delay.

The 3/12 configuration gives a mean of 23.89 hours at €840 per staffed hour, compared with 23.81 hours and €920 at 5/12. It is a candidate for further assessment, not proof of equivalent service: the factorial output does not retain an uncertainty interval for the difference, and no service-level target or budget was specified. BpmnGo's selected 5/15 configuration minimizes mean cycle among the tested points; it is not a demonstrated economic optimum.

3.4 Metric limitations

Every admitted case is drained to completion. Consequently, a completed campaign does not imply a stable queue: the overloaded baseline is a finite-batch result, not a steady-state waiting-time estimate. Throughput divides completions by elapsed time through the final completion, and utilization uses the same extended period. This explains why adequately staffed runs can report fewer than four completions per hour. Counts are integer averages per replication, approximately 2869 cases, not totals across all replications. Reported p95 values average per-replication percentiles. Confidence intervals use $\bar{x} \pm 1.96s/\sqrt{20}$.

Labor cost per case remains around €177 because staffing mainly changes waiting, not the amount of work. It excludes idle payroll. The separate budget measure is $40c_{L1} + 60c_{L2}$ per hour. Equal initial seeds improve reproducibility but do not ensure identical case paths across configurations: changes in event order alter random-draw consumption. Small differences in costs and paths should therefore not be interpreted as redesign benefits.

4 Exercise 4: issue register and redesign proposals

The register in Table 4 distinguishes assignment assumptions from simulation measurements. Severity and proposed responsibility are analyst judgments. All six issues remain open: proposed interventions have not

been deployed. The fixed internal delays of Exercise 1 must not be counted again as gains in Exercise 3, where they were explicitly removed.

Table 4: Issue register.

ID	Issue and evidence	Consequence / proposed owner	Severity
I1	Level-2 overload: workload is 10.4 staff-hours/hour; baseline mean queue is 1166.92 requests.	Unstable capacity at the original demand. Service desk manager.	High
I2	Internal pickup delays: 1 h before checking, 2 h before evaluation, 20 h before resolution and 20 h before delivery.	Expected contribution of 42.6 h per case in Exercise 1. Workflow owner.	High
I3	Level-1 relays a written resolution (20 min) and forwards failed feedback (2 min).	Additional labor and transfers. Process owner.	Medium
I4	Only 20% are known at Level-1; 80% are escalated.	3.2 new Level-2 requests/hour before retries. Knowledge/training lead.	Medium
I5	Each resolution fails with probability 20%.	1.25 attempts per escalated request on average. Level-2 quality lead.	High
I6	Client feedback takes 20 h per delivered resolution.	About 20 h of expected cycle time per initial request. Client-service owner.	Medium

The following proposals apply the redesign heuristic terminology of Reijers and Limam Mansar [2]. Their helpdesk-specific justification and test plans are developed here.

C1 — Extra resources (I1). Validate sufficient qualified Level-2 capacity, using 12 as a candidate under the original workload. The measured 5/12 result directly addresses queue growth, but raises the hourly staffing budget. Reassess capacity after changing work or routing; additional employees should not substitute for analyzing unnecessary work.

C2 — Task automation (I2). Introduce event-driven assignment and notification to reduce avoidable pickup and polling delay. First measure dispatch latency separately from capacity waiting: the assignment does not identify how much of each stated delay is attributable to batch handling rather than staff shortage. Use a separate timing sensitivity study for the original fixed delays; an alert cannot remove a genuine resource queue.

C3 — Task elimination and task automation (I3). Deliver prepared resolutions automatically and route unsuccessful client feedback directly to Level-2. Retain human assistance for cases requiring explanation. This targets duplicate relaying rather than diagnosis. Test labor and cycle effects at unchanged staffing and probabilities, and account for automation costs before claiming financial savings.

C4 — Specialist–generalist (I4). Broaden Level-1 competence through targeted training and a maintained knowledge base, retaining escalation for difficult cases. Test known/unknown probabilities of 0.2/0.8, 0.3/0.7 and 0.4/0.6 as paired scenarios. With other assumptions fixed, a 40% known rate would reduce Level-2 workload to 7.8 staff-hours/hour. This is a conditional calculation, not an observed improvement; training and additional checking effort, and the risk of incorrect closure, must be included.

C5 — Control addition (I5). Add proportionate output validation before delivery. Test check durations of 0, 5 and 10 minutes together with explicitly hypothesized failure rates of 0.20, 0.15 and 0.10. Reduced rework may offset the extra effort, but checking can also worsen an overloaded Level-2 queue. Validate quality and total labor, rather than lowering the failure probability without paying for the proposed control.

C6 — Integration and task automation (I6). Provide direct test-result submission, clear instructions and proportionate reminders. Test 20-, 10- and 5-hour client-response delays as sensitivity scenarios, then measure actual behavior in a pilot. Client availability remains external to the desk; silence must not be treated as successful resolution.

A practical sequence is to validate capacity, instrument dispatch, simplify the relay, then pilot skill, quality and feedback improvements. C3 and C6 can share a communication channel. Staffing must be recalculated after interventions change workload. Six detailed entries, with evidence, heuristics, justification, trade-offs and validation plans, were persisted in BpmnGo's issue register.

5 Exercise 5: proposed to-be model and evaluation

5.1 Selected changes

Figure 1 implements a subset of C2, C3 and C6. The model preserves request checking, the known-resolution branch, initial Level-2 evaluation, problem solving and writing. A new exclusive gateway selects either automated notification or Level-1 assisted delivery. Both paths merge before receiving the client's test result. Confirmed successes close; failures are routed automatically back to the merge before problem solving. Initial evaluation is not repeated.

The lanes separate Level-1 work, Level-2 work, and workflow automation/client-response handling. The receive task represents the desk waiting for external feedback; the client's internal testing process is outside the diagram. The design was imported into BpmnGo's modeler, exported as editable BPMN, and saved as a separate simulation project. Service-task symbols describe proposed integrations, not already deployed notification or routing services.

5.2 Additional assumptions and matched comparison

The main scenario assumes 90% of post-writing deliveries can be automated and 10% require human assistance. This new probability is illustrative, not supplied by the assignment. The assistance flag is assumed available without an additional assessment task and is sampled at each delivery attempt. The assisted path retains the original normal communication duration (mean 20 min, standard deviation 5 min). Both automated tasks take one minute, require no human resource, and have zero modeled incremental cost; development and operating costs remain unestimated.

Known/unknown probabilities remain 0.2/0.8, solved/rework probabilities remain 0.8/0.2, and client waiting remains 1200 minutes. No benefit from faster client responses, better first-level resolution or fewer failures is assumed. The primary as-is and to-be runs both use 5/12 staffing, four arrivals per hour, 720 hours, 20 replications and seed 42. Additional runs test 0% and 100% automated delivery at 5/12 and 90% automation at 5/3.

5.3 Simulation results and interpretation

Table 5: As-is/to-be comparison and sensitivity checks. Cycle times are hours.

Design	L1/L2	Auto. delivery (%)	Mean cycle	p95 cycle	Labor €/case
As-is	5/12	–	23.81	48.23	177.36
To-be	5/12	90	23.52	47.79	164.94
To-be	5/3	90	1,054.32	1,998.91	164.26
To-be	5/12	0	23.77	48.40	176.49
To-be	5/12	100	23.50	47.74	163.59

At matched 5/12 staffing, modeled labor cost falls from €177.36 to €164.94 per case, approximately 7.0%. A simple expected-work check for 90% automation is

$$C_{\text{to-be}} = \frac{10 + 0.2(5) + 0.1(20)}{60}(40) + \frac{156}{60}(60) \approx €164.67.$$

The simulation differs slightly because of stochastic routing and nonnegative-normal sampling. Mean cycle changes from 23.81 to 23.52 hours, a difference of 17.0 minutes. The approximate 95% intervals are 23.68–23.93 hours and 23.34–23.71 hours; they overlap. A separate uncertainty analysis of the difference is required before asserting a statistically reliable cycle-time improvement.

The payroll budget remains €920 per staffed hour in both 5/12 runs. The reduction is in processing labor per case, not demonstrated payroll savings or a full return-on-investment estimate. The zero-automation variant still routes failed feedback automatically, so it is not identical to the original as-is process.

At 5/3, the to-be mean cycle is still approximately 1054 hours. Level-2 work remains about 10.4 staff-hours per incoming hour; automation of the relay cannot resolve that capacity shortage. Once adequate capacity is

available, the retained 20-hour client delay limits further elapsed-time improvement. The proposed design therefore addresses avoidable relay work while preserving the need for a separate capacity decision and evidence about automation coverage.

Overall assessment

The original process combines a capacity bottleneck, repeated work and substantial external waiting. Staffing changes have a large effect where capacity is inadequate; process simplification reduces routine labor but does not replace specialist work. The selected to-be model is a coherent proposal with explicit exceptions and unchanged outcome probabilities. Its operational adoption would require validation of automation coverage, communication quality, service targets and implementation costs. The issue register remains open because simulation is evidence for a design proposal, not confirmation of a production improvement.

A Reproducibility and supporting files

The local BpmnGo campaigns retain their input snapshots and run identifiers [3]. The source package includes the following evidence in `data/`:

Exercise	Supporting files
1	`exercise1-analysis.json`
2	`exercise2-resolution.json`, `exercise2-full-visit.json`
3	`exercise3-comparison.csv`, `exercise3-summary.json`
4	`exercise4-register.json`
5	`exercise5-comparison.csv`, `exercise5-summary.json`

The editable model is `helpdesk-to-be.bpmn`. All five Exercise 5 runs completed with admitted and completed counts matching; the unchanged 5/12 run reproduced Exercise 3's mean cycle and labor cost. Final model references and parameters were checked. Reproduction requires the same engine behavior as well as the same initial seed. Compile this report twice with pdfLaTeX, keeping the `figures/` directory beside `report.tex`.

References

[1] INFO-H420 course materials. *Assignment 2: Business Process Analysis and Redesign*. Provided assignment statement and accompanying BPMN model.

[2] H. A. Reijers and S. Limam Mansar. "Best practices in business process redesign: an overview and qualitative evaluation of successful redesign heuristics." *Omega*, 33(4):283–306, 2005. doi:10.1016/j.omega.2004.04.012.

[3] BpmnGo local simulation laboratory. Saved scenario snapshots, analytical outputs, issue register and simulation campaigns for this assignment. Numerical evidence is included in the accompanying data files.

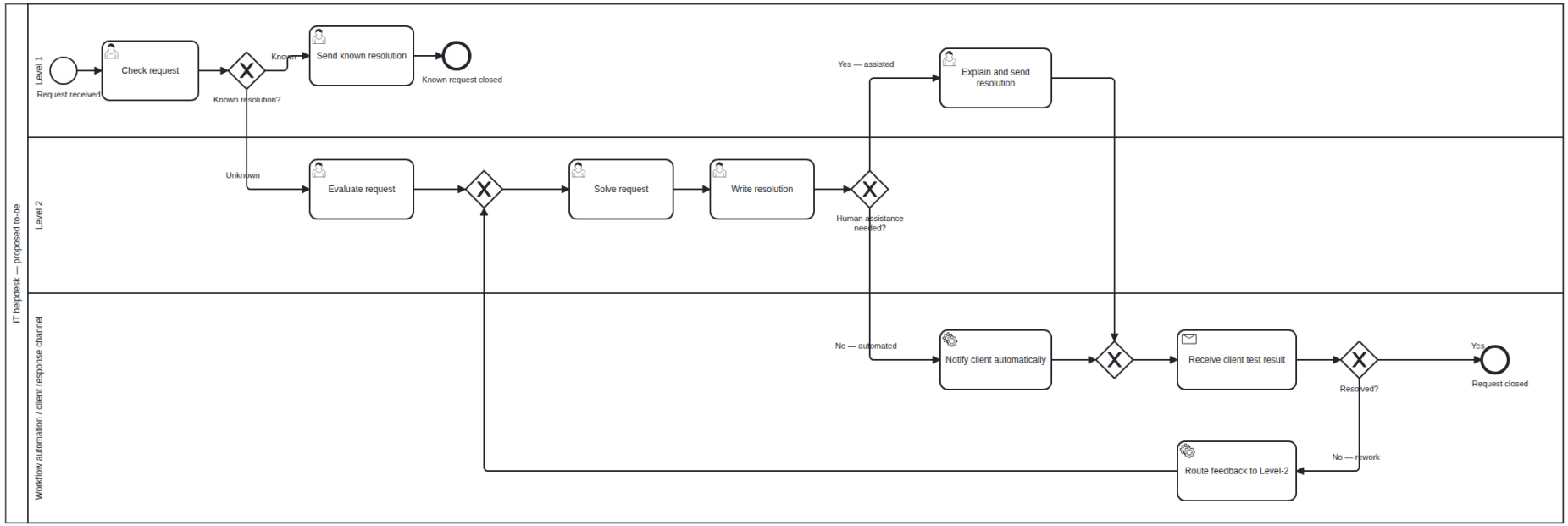


Figure 1: Proposed to-be helpdesk process. Routine delivery and unsuccessful-feedback routing are automated, while a human-assisted delivery branch remains available. The original success and escalation probabilities are preserved.